A Primer on Entropic Optimal Transport and the Schrödinger Bridge

Optimal Transport / Schrödinger Bridge Project
Companion to `python/shared/mixture_sb.py` and `paper_route_sb.py`

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Abstract

This note builds, from scratch, the probabilistic vocabulary needed to read Gushchin et al., *Building the Bridge of Schrödinger* (NeurIPS 2023, arXiv:2306.10161), Sections 2–3, and to see why our closed-form mixture reference is exact. Each section introduces only what the next one needs: measures and random variables, couplings and conditional distributions, entropy, the three optimal transport problems (Monge, Kantorovich, entropic), weak OT, Brownian motion and the heat kernel, the Schrödinger bridge and its two potentials, and finally the "choose the potential" construction. Worked examples are one-dimensional Gaussians, where every object is a number. Boxes titled *Reading Gushchin et al.* say which paragraph of the paper each piece unlocks.

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1 Conventions used throughout

Three quantities are written differently in every source. Fix them once.

- The noise level of the stochastic process is σ^2: the reference process is $dX_t = \sigma\, dW_t$, i.e. Brownian motion with variance $\sigma^2 t$ per coordinate.
- Our solver's constraint is $\partial_t \rho + \nabla \cdot m = \varepsilon \Delta \rho$. Its Fokker–Planck equation (Section 7) forces
$$\varepsilon = \tfrac{1}{2}\sigma^2.$$
- Gushchin et al. call σ^2 simply ϵ; Bunne et al. (2023) call it σ^2; Janati et al. (2020) build it into their regularization constant. So *every ϵ in Gushchin's paper is 2ε in our code.* Section 11 has the full table.

The spatial domain is $\mathbb{R}^d$ in the theory and $[0,1]^d$ with no-flux walls in the solver; Section 10 says when the difference matters. The transport cost is always $c(x,y) = \frac{1}{2}|x-y|^2$.

2 Measures, densities, random variables

Definition 2.1 (Probability measure, density)**.** A probability measure μ on $\mathbb{R}^d$ assigns to every (measurable) set A a mass $\mu(A) \in [0,1]$ with $\mu(\mathbb{R}^d) = 1$. When $\mu(A) = \int_A \mu(x)\, dx$ for a nonnegative function $\mu(x)$ we say μ has a density, and we use the same letter for the measure and the density. $\mathcal{P}(\mathbb{R}^d)$ is the set of probability measures; $\mathcal{P}_2(\mathbb{R}^d)$ those with finite second moment $\int |x|^2 d\mu < \infty$.

Everything our grid solver sees is a density (ρ_0, ρ_1, ρ_t), but the theory needs two measures without densities: the Dirac mass δ_a (all mass at the point a), and couplings concentrated on a curve (Section 3).

Definition 2.2 (Random variable, law, expectation)**.** A random variable X with values in $\mathbb{R}^d$ has a *law* (or distribution) μ, written $X \sim \mu$, meaning $\mathbb{P}(X \in A) = \mu(A)$. The expectation of a function of X is
$$\mathbb{E}[f(X)] = \int f\, d\mu = \int f(x)\, \mu(x)\, dx.$$

The notations $\int f\, d\mu$, $\int f(x)\, d\mu(x)$, $\int f(x)\, \mu(dx)$ and $\mathbb{E}_{X\sim\mu}[f(X)]$ all mean the same thing. A random variable is just a convenient name for "a point drawn from μ"; nothing is random in the formulas, which are all integrals.

Definition 2.3 (Pushforward)**.** If $X \sim \mu$ and $T : \mathbb{R}^d \to \mathbb{R}^d$, the law of $T(X)$ is the pushforward $T_\# \mu$, defined by $(T_\#\mu)(A) = \mu(T^{-1}(A))$. For a smooth invertible T and densities, the change of variables formula reads $(T_\#\mu)(T(x))\, |\det \nabla T(x)| = \mu(x)$.

Definition 2.4 (Gaussians)**.** $\mathcal{N}(m, \Sigma)$ is the Gaussian with mean m and covariance Σ; its density at x is written $\mathcal{N}(x; m, \Sigma)$. In 1D, $\mathcal{N}(x; m, s^2) = (2\pi s^2)^{-1/2} e^{-(x-m)^2/2s^2}$.

Two Gaussian identities do most of the work later. Both hold in any dimension d, with A, B symmetric positive-definite $d \times d$ covariance matrices. First, the *product* of two Gaussian densities in the same variable is a Gaussian density times a constant:
$$\mathcal{N}(y; a, A)\, \mathcal{N}(y; b, B) = \mathcal{N}(a; b, A+B)\, \mathcal{N}(y; m, S), \quad S = (A^{-1}+B^{-1})^{-1}, \;\; m = S\,(A^{-1}a + B^{-1}b). \tag{1}$$

The new covariance adds *precisions* (inverse covariances), and the new mean is the precision-weighted average of the two means. In 1D, with $A = \alpha$, $B = \beta$ scalars, this reads $S = \alpha\beta/(\alpha+\beta)$ and $m = (\beta a + \alpha b)/(\alpha + \beta)$. The constant $\mathcal{N}(a; b, A + B)$ is the integral of the left-hand side over y, so it is also an instance of the second identity. Second, the *convolution* of two Gaussians adds covariances:

$$\int \mathcal{N}(y; x, A)\,\mathcal{N}(x; m, B)\,dx = \mathcal{N}(y; m, A + B). \tag{2}$$

Probabilistically, (2) says: if $X \sim \mathcal{N}(m, B)$ and, given $X = x$, $Y \sim \mathcal{N}(x, A)$, then $Y \sim \mathcal{N}(m, A + B)$. "Adding independent Gaussian noise adds variance."

Definition 2.5 (Product measure)**.** For $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$, the product $\mu \otimes \nu$ is the measure on $\mathbb{R}^d \times \mathbb{R}^d$ with density $\mu(x)\nu(y)$. It is the law of a pair (X, Y) with $X \sim \mu$, $Y \sim \nu$ and X, Y *independent*.

Reading Gushchin et al.

Gushchin's Section 2 opens with $\mathcal{P}_2(\mathbb{R}^D)$ and calls the two marginals P and Q. Our ρ_0 is their P, our ρ_1 is their Q. Measures that are "absolutely continuous" are the ones with a density.

3 Couplings and conditional distributions

This is the section the rest of the primer stands on.

Definition 3.1 (Joint law, marginals, coupling)**.** Let (X, Y) be a pair of random vectors. Its *joint law* $\pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ has density $\pi(x, y)$. The *marginals* are the laws of X and Y alone,

$$\pi_0(x) = \int \pi(x, y)\,dy, \qquad \pi_1(y) = \int \pi(x, y)\,dx.$$

π is a *coupling* (or transport plan) of ρ_0 and ρ_1 if $\pi_0 = \rho_0$ and $\pi_1 = \rho_1$. The set of all couplings is $\Pi(\rho_0, \rho_1)$.

$\Pi(\rho_0, \rho_1)$ is never empty: the product $\rho_0 \otimes \rho_1$ is in it. At the other extreme, if $T_{\#}\rho_0 = \rho_1$, the pair $(X, T(X))$ with $X \sim \rho_0$ is a coupling too. It has no density on $\mathbb{R}^d \times \mathbb{R}^d$: all its mass sits on the graph $\{y = T(x)\}$. Such a coupling is called *deterministic*, and it is what the Monge problem optimizes over (Section 5). Think of a coupling as a recipe "take a particle at x and send it to y", which may split the mass at x among several y.

Definition 3.2 (Conditional distribution)**.** For a coupling with density, the *conditional distribution of Y given $X = x$* is the probability measure on $\mathbb{R}^d$ with density

$$\pi(y \,|\, x) = \frac{\pi(x, y)}{\rho_0(x)}. \tag{3}$$

It is written $\pi(\cdot \,|\, x) \in \mathcal{P}(\mathbb{R}^d)$: for each fixed x, a probability measure in the variable y.

Equation (3) is a definition, not a theorem; it is the slice of the joint density at x, renormalized to unit mass. It is exactly the "where does the mass at x go" recipe above. Three consequences, each used later:

1. **Disintegration.** $\pi(x, y) = \rho_0(x)\,\pi(y|x)$. The joint law is the first marginal times the conditional. Books write this as $\pi(dx, dy) = \rho_0(dx)\,\pi(dy|x)$, or $\pi = \rho_0 \otimes \pi(\cdot|x)$, which also covers the case without densities.

2. **Second marginal from the first.**

$$\rho_1(y) = \int \rho_0(x)\,\pi(y \mid x)\,dx. \tag{4}$$

   If you know where each x sends its mass, the target is determined. This one line is the entire idea of Gushchin's construction: choose ρ_0 and the conditionals, read off ρ_1.

3. **Conditional expectation.** $\mathbb{E}[Y \mid X = x] = \int y\,\pi(y|x)\,dy$, the mean destination of the mass at x. The map $x \mapsto \mathbb{E}[Y|X = x]$ is called the barycentric projection of the plan.

A family $x \mapsto \pi(\cdot|x)$ of probability measures is also called a *Markov kernel* or *transition kernel*; the words mean "a conditional distribution for every x".

Example 3.3 (Bivariate Gaussian)**.** Let (X, Y) be jointly Gaussian in 1D with means μ_0, μ_1, variances a^2, b^2 and covariance $c = \mathbb{E}[(X - \mu_0)(Y - \mu_1)]$. Completing the square in the joint density gives

$$\pi(\cdot \mid x) = \mathcal{N}\Big(\mu_1 + \frac{c}{a^2}(x - \mu_0),\quad b^2 - \frac{c^2}{a^2}\Big). \tag{5}$$

The mean destination is affine in x; the spread of the destinations is the same for every x. Two limits: $c = ab$ gives zero conditional variance, so $Y = \mu_1 + \frac{b}{a}(X - \mu_0)$ exactly (a deterministic coupling, the Monge map between the two Gaussians); $c = 0$ gives $\pi(\cdot|x) = \mathcal{N}(\mu_1, b^2)$ for every x, the product coupling. The entropic plan between Gaussians (Section 5) is this family with $c = C_\sigma$ from Bunne et al., sliding from ab at $\sigma = 0$ to 0 as $\sigma \to \infty$.

Reading Gushchin et al.

Gushchin writes a plan as $\pi \in \Pi(P, Q)$ and its conditionals as $\pi(\cdot|x)$, exactly as here. Whenever the paper says "π is a mixture of $\pi(\cdot|x)$ over $x \sim P$" it means the disintegration above. Their sampling procedure for Q is (4) read as an algorithm: draw $x \sim P$, then $y \sim \pi(\cdot|x)$.

4 Entropy and Kullback–Leibler divergence

Definition 4.1 (Differential entropy)**.** For a density μ, $H(\mu) = -\int \mu \log \mu\,dx$. For a Gaussian, $H(\mathcal{N}(m, \Sigma)) = \frac{1}{2}\log\big((2\pi e)^d \det \Sigma\big)$; in 1D, $H = \frac{1}{2}\log(2\pi e\, s^2)$.

Entropy is large for spread-out densities and goes to $-\infty$ as a density concentrates on a point. This is the mechanism by which an entropy term forbids deterministic couplings.

Definition 4.2 (KL divergence)**.** For probability measures μ, ν with densities,

$$\mathrm{KL}(\mu \,\|\, \nu) = \int \mu(x) \log \frac{\mu(x)}{\nu(x)}\,dx.$$

It is ≥ 0 with equality only when $\mu = \nu$ (Jensen's inequality), it is not symmetric, and it is $+\infty$ when μ puts mass where ν has none (then μ is not "absolutely continuous" with respect to ν, written $\mu \not\ll \nu$).

In 1D with equal means, $\mathrm{KL}(\mathcal{N}(0,a^2)\|\mathcal{N}(0,b^2)) = \frac{1}{2}\big(\frac{a^2}{b^2} - 1 - \log\frac{a^2}{b^2}\big)$. Note $\mathrm{KL}(\mu\|\nu) = -H(\mu) - \int \mu \log \nu$: KL is negative entropy plus a cross term.

Lemma 4.3 (KL splits along conditionals)**.** *Let π and ν be two measures on $\mathbb{R}^d \times \mathbb{R}^d$ with the same first marginal ρ_0. Then*

$$\mathrm{KL}(\pi \,\|\, \nu) = \int \rho_0(x)\ \mathrm{KL}\left(\pi(\cdot|x) \,\big\|\, \nu(\cdot|x)\right) dx.$$

Proof. Write $\pi(x,y) = \rho_0(x)\pi(y|x)$ and $\nu(x,y) = \rho_0(x)\nu(y|x)$ inside the logarithm; the ρ_0 factors cancel. □

The case $\nu = \rho_0 \otimes \rho_1$, the independent coupling, is the one entropic OT uses:

$$\mathrm{KL}(\pi \,\|\, \rho_0 \otimes \rho_1) = \int \rho_0(x)\ \mathrm{KL}\left(\pi(\cdot|x) \,\|\, \rho_1\right) dx = H(\rho_1) - \int \rho_0(x)\, H\big(\pi(\cdot|x)\big)\, dx. \tag{6}$$

This is the *mutual information* of (X,Y): zero for independent variables, infinite for a deterministic coupling (each $\pi(\cdot|x)$ is a Dirac, entropy $-\infty$). For the bivariate Gaussian of Example 3.3 with correlation $r = c/(ab)$ it equals $-\frac{1}{2}\log(1-r^2)$. The second form in (6) is what makes entropic OT a "weak" problem in Section 6: for fixed marginals, the only part that depends on the plan is the average entropy of the conditionals.

5 Monge, Kantorovich and entropic optimal transport

Definition 5.1 (Monge)**.** Given ρ_0, ρ_1, minimize $\int \frac{1}{2}|x - T(x)|^2 \rho_0(x)\, dx$ over maps T with $T_{\#}\rho_0 = \rho_1$.

Definition 5.2 (Kantorovich)**.** Minimize over couplings instead of maps:

$$\tfrac{1}{2}W_2^2(\rho_0,\rho_1) = \min_{\pi \in \Pi(\rho_0,\rho_1)} \iint \tfrac{1}{2}|x-y|^2\, \pi(x,y)\, dx\, dy = \min_{\pi}\ \mathbb{E}_{(X,Y)\sim\pi} \tfrac{1}{2}|X-Y|^2.$$

Every map gives a coupling, so Kantorovich's value is at most Monge's; it is a relaxation (a "weaker" problem in the sense of fewer constraints). Brenier's theorem says that for ρ_0 with a density the Kantorovich optimum is deterministic, $y = \nabla\psi(x)$ with ψ convex, so the two values agree. For Gaussians, $T(x) = \mu_1 + A(x-\mu_0)$ with $A = \Sigma_0^{-1/2}(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2})^{1/2}\Sigma_0^{-1/2}$ and $W_2^2 = |\mu_0-\mu_1|^2 + \operatorname{Tr}\Sigma_0 + \operatorname{Tr}\Sigma_1 - 2\operatorname{Tr}(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2})^{1/2}$, the Bures–Wasserstein distance (Janati et al., Section 2).

Definition 5.3 (Entropic OT)**.** For $\sigma^2 > 0$,

$$\mathrm{OT}_{\sigma^2}(\rho_0,\rho_1) = \min_{\pi \in \Pi(\rho_0,\rho_1)} \iint \tfrac{1}{2}|x-y|^2\, d\pi + \sigma^2\ \mathrm{KL}(\pi \,\|\, \rho_0 \otimes \rho_1). \tag{7}$$

The penalty rules out deterministic plans (infinite KL) and pulls the plan toward independence as σ^2 grows. Replacing $\mathrm{KL}(\pi\|\rho_0\otimes\rho_1)$ by $-H(\pi)$, or by $\mathrm{KL}(\pi\|\text{Lebesgue})$, changes the objective by a constant that depends only on ρ_0, ρ_1, so all three conventions have the same minimizer. The regularization constant depends on how the cost is scaled: with $\frac{1}{2}|x-y|^2$ it is σ^2 as in (7), with $|x-y|^2$ it is $2\sigma^2$ (Janati's convention). Both are chosen so that the entropic plan is the one of the Schrödinger bridge with noise σ^2 (Section 8).

Proposition 5.4 (Form of the entropic plan)**.** *The minimizer of* (7) *is unique and has the form*

$$\pi^*(x,y) = \hat\varphi(x)\,\mathcal{K}_\sigma(x,y)\,\varphi(y), \qquad \mathcal{K}_\sigma(x,y) = \exp\Big(-\frac{|x-y|^2}{2\sigma^2}\Big), \tag{8}$$

for two nonnegative functions $\hat\varphi, \varphi$*, the* Schrödinger potentials*, which are determined (up to* $\hat\varphi \to \lambda\hat\varphi$*,* $\varphi \to \varphi/\lambda$*) by the two marginal constraints, the* Schrödinger system

$$\rho_0(x) = \hat\varphi(x)\int \mathcal{K}_\sigma(x,y)\varphi(y)\,dy, \qquad \rho_1(y) = \varphi(y)\int \hat\varphi(x)\mathcal{K}_\sigma(x,y)\,dx. \tag{9}$$

Conversely, any π *of the form* (8) *that lies in* $\Pi(\rho_0,\rho_1)$ *is the minimizer.*

Where it comes from: the Lagrangian of (7) with multipliers $f(x), g(y)$ for the two marginal constraints is minimized by $\pi^* = \exp\big((f(x)+g(y)-\frac12|x-y|^2)/\sigma^2\big)\rho_0(x)\rho_1(y)$; set $\hat\varphi = \rho_0 e^{f/\sigma^2}$, $\varphi = \rho_1 e^{g/\sigma^2}$. Sinkhorn's algorithm is the alternating solution of (9) for $\hat\varphi$ and φ. The converse is the key to Gushchin: *the exponential form plus the right marginals is a certificate of optimality*, because the problem is strictly convex and the form is exactly the first-order condition.

The conditional distributions of the entropic plan follow from (3) and (9):

$$\pi^*(y\,|\,x) = \frac{\mathcal{K}_\sigma(x,y)\,\varphi(y)}{\int \mathcal{K}_\sigma(x,y')\varphi(y')\,dy'}. \tag{10}$$

Read (10) as: *a Gaussian of variance* σ^2 *centered at* x*, reweighted by* φ*, renormalized.* The mass at x spreads out by diffusion and is then attracted to where φ is large. Only the terminal potential φ enters; $\hat\varphi$ is fixed afterwards by ρ_0. This is the sentence to hold on to while reading Section 3 of the paper.

Example 5.5 (Gaussians)**.** If φ is the exponential of a quadratic, so is $\mathcal{K}_\sigma\varphi$, and the conditional (10) is Gaussian with a mean affine in x: the plan is a bivariate Gaussian, i.e. Example 3.3. Solving (9) for the covariance c in 1D gives the quadratic $c^2 + \sigma^2 c = a^2b^2$, hence $c = C_\sigma = \frac12\big(\sqrt{4a^2b^2+\sigma^4} - \sigma^2\big)$. The matrices D_σ and C_σ of Bunne et al. are the d-dimensional version of this quadratic formula.

Reading Gushchin et al.

Section 2 of the paper states entropic OT, remarks that it has a unique minimizer, and introduces the potentials. Their ϵ is our σ^2. If a formula there carries $\exp(-|x-y|^2/2\epsilon)$, that is $\mathcal{K}_\sigma$.

6 Weak optimal transport

Kantorovich's cost is a sum over pairs: $\int c\,d\pi = \int \rho_0(x)\big[\int c(x,y)\pi(y|x)dy\big]dx$. The bracket is *linear* in the conditional $\pi(\cdot|x)$. Weak OT (Gozlan, Roberto, Samson, Tetali 2017; Backhoff-Veraguas, Beiglböck, Pammer 2019) allows a nonlinear dependence.

Definition 6.1 (Weak OT)**.** Let $C : \mathbb{R}^d \times \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ be a cost of a point and a probability measure, convex in its second argument. The weak OT problem is

$$\min_{\pi\in\Pi(\rho_0,\rho_1)} \int \rho_0(x)\,C\big(x,\pi(\cdot|x)\big)\,dx. \tag{11}$$

If $C(x,\mu) = \int c(x,y)\,d\mu(y)$, this is Kantorovich with cost c; nothing changes. The word "weak" refers to the cost seeing the conditional law only through integrals ("weakly"), not to a weakening of constraints. It is a generalization of Kantorovich, whereas Kantorovich is a relaxation of Monge.

Proposition 6.2 (Entropic OT is weak OT)**.** *By (6), problem (7) equals, up to the constant $\sigma^2 H(\rho_1)$, problem (11) with*

$$C(x,\mu) = \int \tfrac{1}{2}|x-y|^2\,d\mu(y) - \sigma^2 H(\mu). \tag{12}$$

The nonlinear part is the entropy of the conditional: the plan pays for sending the mass at x to a sharply defined destination.

Lemma 6.3 (Gibbs variational principle)**.** *For a function V with $\int e^{-V/\sigma^2} < \infty$,*

$$\min_{\mu\in\mathcal{P}(\mathbb{R}^d)} \Big\{ \int V\,d\mu - \sigma^2 H(\mu) \Big\} \quad \textit{is attained only at} \quad \mu^*(y) = \frac{e^{-V(y)/\sigma^2}}{\int e^{-V/\sigma^2}}.$$

Proof. $\int V d\mu - \sigma^2 H(\mu) = \sigma^2\,\mathrm{KL}(\mu\|\mu^*) - \sigma^2 \log \int e^{-V/\sigma^2}$, and $\mathrm{KL} \geq 0$ with equality only at $\mu = \mu^*$. □

Weak OT has a dual problem with a single potential f on the target space, and its optimality condition is pointwise in x:

$$\pi^*(\cdot\,|\,x) = \arg\min_{\mu\in\mathcal{P}(\mathbb{R}^d)} \Big\{ C(x,\mu) - \int f\,d\mu \Big\} \quad \text{for } \rho_0\text{-almost every } x. \tag{13}$$

For the entropic cost (12) this is Lemma 6.3 with $V(y) = \frac{1}{2}|x-y|^2 - f(y)$, giving

$$\pi^*(y|x) \;\propto\; e^{-|x-y|^2/2\sigma^2}\, e^{f(y)/\sigma^2} = \mathcal{K}_\sigma(x,y)\,\varphi(y), \qquad \varphi := e^{f/\sigma^2},$$

which is (10) again. The dual potential of weak OT and the Schrödinger potential are the same object, $\varphi = e^{f/\sigma^2}$. The marginal constraint $\rho_1 = \int \rho_0\,\pi^*(\cdot|x)\,dx$ is the equation that determines f when both marginals are given. Run backwards, it says: *pick f freely, and ρ_1 is whatever (4) produces.*

Reading Gushchin et al.

The weak-OT paragraph of Section 2 is there only to state (13): the optimal conditional plans are the minimizers of a pointwise problem driven by one potential. If the notation $C(x,\mu)$ with a measure as second argument is unfamiliar, it is (12).

7 Brownian motion, SDEs, the heat kernel

Definition 7.1 (Brownian motion)**.** W_t, $t \geq 0$, is a Brownian motion in $\mathbb{R}^d$ if $W_0 = 0$, its paths are continuous, and its increments over disjoint intervals are independent with $W_t - W_s \sim \mathcal{N}(0,(t-s)I)$.

The process $X_t = x + \sigma W_t$ ("Brownian motion with volatility σ", written $dX_t = \sigma\, dW_t$) has $X_t \sim \mathcal{N}(x, \sigma^2 t I)$. Its *transition density*

$$p_t(x, y) = \mathcal{N}(y;\, x,\, \sigma^2 t I) = (2\pi\sigma^2 t)^{-d/2} \exp\Big(- \frac{|x-y|^2}{2\sigma^2 t} \Big) \tag{14}$$

is the probability density of being at y at time t having started at x. In y it solves the heat equation $\partial_t p = \frac{\sigma^2}{2}\Delta p$ with $p_0 = \delta_x$; the same equation holds in x by symmetry. The Gaussian convolution rule (2) is the Chapman–Kolmogorov identity $p_{s+t} = \int p_s(x, z) p_t(z, y)\, dz$.

At $t = 1$, $p_1(x, y)$ is $\mathcal{K}_\sigma(x, y)$ of (8) up to the constant $(2\pi\sigma^2)^{-d/2}$. *The entropic kernel is the one-step heat kernel.* This is the whole reason entropic OT and the Schrödinger bridge are the same problem.

Definition 7.2 (SDE with drift)**.** $dX_t = u(t, X_t)\, dt + \sigma\, dW_t$ describes a particle pushed by the velocity field u and jostled by noise. If $X_0 \sim \rho_0$, the law ρ_t of X_t solves the Fokker–Planck equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t u) = \tfrac{\sigma^2}{2}\Delta \rho_t. \tag{15}$$

With $m = \rho u$ this is our solver's constraint $\partial_t \rho + \nabla \cdot m = \varepsilon \Delta \rho$ with $\varepsilon = \sigma^2/2$. For $u = 0$ it is the heat equation.

Definition 7.3 (Path measure, Brownian bridge)**.** The law of the whole trajectory $(X_t)_{t\in[0,1]}$ is a probability measure on the space of continuous paths; for $dX = \sigma dW$ started from ρ_0 it is the *Wiener measure* $\mathbb{W}^\sigma$. Conditioning Brownian motion on $X_0 = x$ and $X_1 = y$ gives the *Brownian bridge*, a Gaussian process with

$$X_t \sim \mathcal{N}\big((1-t)x + ty,\ \sigma^2 t(1-t) I\big).$$

The bridge interpolates linearly in the mean and bulges in the middle; the term $\sigma^2 t(1-t)I$ in Bunne's Σ_t formula is this bulge.

Reading Gushchin et al.

Gushchin's reference process is Brownian motion with volatility ϵ, i.e. our $\sigma^2 = \epsilon$ (equivalently $dX = \sqrt{\epsilon}\, dW$). The joint law of (X_0, X_1) under it has density $\rho_0(x)\, p_1(x, y)$, which they write as $P\otimes$ the heat kernel.

8 The Schrödinger bridge

Definition 8.1 (Dynamic Schrödinger bridge)**.** Among all path measures $\mathbb{P}$ whose time-0 and time-1 marginals are ρ_0 and ρ_1, find the one closest to the Wiener measure:

$$\min_{\mathbb{P}:\ \mathbb{P}_0=\rho_0,\ \mathbb{P}_1=\rho_1} \mathrm{KL}(\mathbb{P} \,\|\, \mathbb{W}^\sigma).$$

Interpretation: Brownian particles start distributed as ρ_0; we observe that at time 1 they are distributed as ρ_1, which is not what diffusion alone would give; the bridge is the most likely way this happened. Two reformulations make it computable.

Control form (what our solver computes). If $\mathbb{P}$ is the law of $dX = u\,dt + \sigma\,dW$ with $X_0 \sim \rho_0$, Girsanov's theorem gives

$$\mathrm{KL}(\mathbb{P}\,\|\,\mathbb{W}^\sigma) = \frac{1}{2\sigma^2}\,\mathbb{E}\int_0^1 |u(t,X_t)|^2\,dt = \frac{1}{\sigma^2}\int_0^1\!\!\int \frac{|m|^2}{2\rho_t}\,dx\,dt, \qquad m = \rho_t u. \tag{16}$$

Minimizing the right-hand side subject to the Fokker–Planck constraint (15) with ρ_0, ρ_1 prescribed is the Benamou–Brenier problem with diffusion,

$$\min_{\rho,m}\int_0^1\!\!\int \frac{|m|^2}{2\rho}\,dx\,dt \quad \text{s.t.} \quad \partial_t\rho + \nabla\cdot m = \varepsilon\Delta\rho, \quad \varepsilon = \tfrac{1}{2}\sigma^2, \tag{17}$$

which is exactly what the LADMM solver discretizes. The optimal u is the *current* velocity of the bridge, and m is its momentum.

Static reduction (what the closed forms use). Any path measure is its joint endpoint law $\pi(x,y)$ times the conditional law of the path given the endpoints. Applying Lemma 4.3 on path space,

$$\mathrm{KL}(\mathbb{P}\|\mathbb{W}^\sigma) = \mathrm{KL}(\pi\,\|\,\rho_0\otimes p_1) + \int \pi(x,y)\ \mathrm{KL}\left(\mathbb{P}(\cdot|x,y)\,\|\,\mathrm{bridge}_{x\to y}\right)dx\,dy,$$

where $\rho_0\otimes p_1$ denotes the density $\rho_0(x)p_1(x,y)$ of the endpoints under Wiener measure. The second term is killed by choosing $\mathbb{P}(\cdot|x,y)$ to be the Brownian bridge, and the first term equals, up to a constant, $\sigma^{-2}\big[\int \frac{1}{2}|x-y|^2 d\pi + \sigma^2\,\mathrm{KL}(\pi\|\rho_0\otimes\rho_1)\big]$, the entropic OT objective (7). Hence:

> The Schrödinger bridge is the entropic OT plan π^* glued to Brownian bridges: draw $(X_0,X_1)\sim\pi^*$, then run a Brownian bridge between them. Its endpoint law is π^*, and its potentials $\hat\varphi, \varphi$ are those of Proposition 5.4.

Potentials in time. Extend the two potentials to all $t\in[0,1]$ by running the heat equation backwards from φ and forwards from $\hat\varphi$:

$$\varphi_t(x) = \int p_{1-t}(x,y)\,\varphi(y)\,dy, \qquad \hat\varphi_t(y) = \int \hat\varphi(x)\,p_t(x,y)\,dx, \tag{18}$$

so $\partial_t\varphi_t = -\frac{\sigma^2}{2}\Delta\varphi_t$ with $\varphi_1 = \varphi$, and $\partial_t\hat\varphi_t = \frac{\sigma^2}{2}\Delta\hat\varphi_t$ with $\hat\varphi_0 = \hat\varphi$.

Proposition 8.2 (Marginals and drift of the bridge)**.** *With $\varphi_t, \hat\varphi_t$ as in (18) (and p_1 in place of $\mathcal{K}_\sigma$ in (8)),*

$$\rho_t = \varphi_t\,\hat\varphi_t, \qquad u(t,x) = \sigma^2\nabla\log\varphi_t(x), \qquad m = \rho_t u = \sigma^2\,\hat\varphi_t\nabla\varphi_t. \tag{19}$$

Check. At $t=0$, $\varphi_0 = \int p_1\varphi$ and $\hat\varphi_0 = \hat\varphi$, so $\rho_0 = \hat\varphi\varphi_0$ is the first Schrödinger equation (9); similarly at $t=1$. For $0<t<1$, differentiate the product: $\partial_t(\varphi_t\hat\varphi_t) = -\frac{\sigma^2}{2}\hat\varphi_t\Delta\varphi_t + \frac{\sigma^2}{2}\varphi_t\Delta\hat\varphi_t$, while $\nabla\cdot(\rho_t u) = \sigma^2\nabla\cdot(\hat\varphi_t\nabla\varphi_t)$ and $\frac{\sigma^2}{2}\Delta(\varphi_t\hat\varphi_t) = \frac{\sigma^2}{2}(\hat\varphi_t\Delta\varphi_t + 2\nabla\varphi_t\cdot\nabla\hat\varphi_t + \varphi_t\Delta\hat\varphi_t)$. Expanding $\nabla\cdot(\hat\varphi_t\nabla\varphi_t) = \nabla\hat\varphi_t\cdot\nabla\varphi_t + \hat\varphi_t\Delta\varphi_t$ shows that (15) holds term by term. So ρ_t is a Fokker–Planck solution with the right endpoints and drift u, and ρ_t is also the time-t marginal of the bridge-glued plan, which is the mixture $\int\pi^*(x,y)\,\mathcal{N}(\cdot;(1-t)x+ty,\sigma^2t(1-t))$. □

The drift is a gradient of a log: Schrödinger bridges are always gradient flows of $\log \varphi_t$, in the same way that Brenier maps are gradients of convex functions. In our convention $m = 2\varepsilon\, \rho_t \nabla \log \varphi_t$.

Reading Gushchin et al.

This is the "dynamic SB" part of Section 2. The paper's φ^* and $\hat{\varphi}^*$ (or $\phi, \hat{\phi}$; check the pdf) are our $\varphi, \hat{\varphi}$; the optimal drift $\epsilon \nabla \log \varphi_t$ is (19) with $\epsilon = \sigma^2$. Their statement that the optimal process "is the reference process with an added drift" is (16).

9 Gushchin's construction: choose the potential

Normally ρ_0, ρ_1 are given and (9) must be solved numerically for $\hat{\varphi}, \varphi$. The construction runs this backwards.

Proposition 9.1 (Constructed pairs, Gushchin et al. Thm. 3.2 in substance)**.** *Fix ρ_0 and any nonnegative φ with $\varphi_0 := \int p_1(\cdot, y)\varphi(y)\, dy$ finite. Define*

$$\hat{\varphi} := \frac{\rho_0}{\varphi_0}, \qquad \pi^*(y|x) := \frac{p_1(x,y)\varphi(y)}{\varphi_0(x)}, \qquad \rho_1(y) := \int \rho_0(x)\, \pi^*(y|x)\, dx = \varphi_1(y)\, \hat{\varphi}_1(y), \tag{20}$$

where $\varphi_1 = \varphi$ by the definition (18), so that the endpoints read $\rho_0 = \varphi_0\hat{\varphi}_0$ and $\rho_1 = \varphi_1\hat{\varphi}_1$, the $t = 0$ and $t = 1$ cases of (19). Then $\pi^*(x, y) = \rho_0(x)\pi^*(y|x) = \hat{\varphi}(x)p_1(x, y)\varphi(y)$ is the entropic plan between ρ_0 and this ρ_1, and the Schrödinger bridge between them has marginals and drift given by (19).

Proof. π^* has the exponential form, its first marginal is ρ_0 by construction (the conditionals are normalized), and its second marginal is ρ_1 by definition. The converse part of Proposition 5.4 does the rest. □

The price is that ρ_1 is an output, not an input. The gain is that the drift $\sigma^2 \nabla \log \varphi_t$ and every ρ_t are exact, with no optimization anywhere.

Gaussian-mixture potentials. Take

$$\varphi(y) = \sum_{k=1}^{K} w_k\, \mathcal{N}(y; \mu_k, \Sigma_k), \qquad w_k > 0. \tag{21}$$

The number of components K and the triples (w_k, μ_k, Σ_k) are *free design parameters*, chosen by hand; nothing in the problem fixes them, and they are not the modes and covariances of ρ_1 (compare $\mu_\pm = \pm a$ with the actual centres $\bar{y}_\pm$ in Example 9.2). They are the input; the marginals are the output. Gushchin pick theirs "to make the constructed distributions visually pleasant" (their Appendix B); ours are the (w, b, Q) and (u, a, P) arguments of `mixture_sb.py`. Gushchin call this an LSE (log-sum-exp) potential because $\log \varphi$ is a log-sum-exp of quadratics. Everything becomes explicit through the two Gaussian identities of Section 2:

- **Time potential.** By the convolution rule (2), $\varphi_t(x) = \sum_k w_k\, \mathcal{N}\big(x; \mu_k, \Sigma_k + \sigma^2(1-t)I\big)$: running the heat equation backwards inflates each component's covariance.

- **Drift.** $u = \sigma^2 \nabla \log \varphi_t$ is a weighted average of the Gaussian gradients,

$$u(t,x) = -\sigma^2 \sum_k \omega_k(t,x)\,\big(\Sigma_k + \sigma^2(1-t)I\big)^{-1}(x-\mu_k),$$
$$\omega_k(t,x) \propto w_k\,\mathcal{N}\big(x;\mu_k,\Sigma_k+\sigma^2(1-t)I\big),$$

  with the responsibilities ω_k summing to one. Each component pulls toward its centre; the pull is strongest where that component dominates.

- **Conditional plan.** $\pi^*(y|x) \propto \varphi(y)\,\mathcal{N}(y;x,\sigma^2 I)$ is a mixture of products of two Gaussians in y, so by the product rule (1) with $A = \Sigma_k$, $B = \sigma^2 I$, i.e. $S_k = (\Sigma_k^{-1} + \sigma^{-2} I)^{-1}$ and $m_k(x) = S_k(\Sigma_k^{-1}\mu_k + \sigma^{-2}x)$,

$$\pi^*(y|x) = \sum_k \omega_k(0,x)\,\mathcal{N}\big(y;m_k(x),S_k\big). \tag{22}$$

  The mass at x is split among the K components with x-dependent weights, and each part lands in a Gaussian around a point between x and μ_k.

- **Target.** $\rho_1 = \int \rho_0(x)\pi^*(\cdot|x)\,dx$ is sampled by $x \sim \rho_0$, then $y \sim \pi^*(\cdot|x)$ (Gushchin's Prop. 3.3). Its density is a d-dimensional integral, closed form only when ρ_0 is itself Gaussian or a Gaussian mixture, which is the case treated next.

Dictionary: the same objects in the paper's notation. Section 3.3 of the paper looks different from (21)–(22) for three reasons: they parametrize the *dual* potential f^* rather than the Schrödinger potential φ; they write Gaussians *unnormalized*, through $Q(y\,|\,b,A) := \exp\big(-\frac{1}{2}(y-b)^\top A\,(y-b)\big)$, a bell of precision A with no $(2\pi)^{-D/2}\det^{-1/2}$ in front; and they parametrize each component by its precision rather than its covariance. To keep one index, we write their component index n as our k, their ϵ as our σ^2, and their weight w_n as c_k, because our w_k is already taken. Two of their names clash with ours and mean different things: their Σ_n and $\mu_n(x)$ are the covariance and mean of the *conditional* plan, our S_k and $m_k(x)$, not the component parameters Σ_k, μ_k of (21).

1. **Dual vs. Schrödinger potential (their (14), Cor. 3.4).** Their conditional plan is $\pi^*(y|x) \propto \exp\big((f^*(y) - \frac{1}{2}|x-y|^2)/\epsilon\big)$. With

$$\varphi(y) = \exp\big(f^*(y)/\epsilon\big)$$

   this is exactly $\mathcal{K}_\sigma(x,y)\varphi(y)$, i.e. our (10). Corollary 3.4 says precisely this: $\exp(f^*/\epsilon)$ is the Schrödinger potential.

2. **The LSE potential (their (15)).** $f^*(y) = \epsilon \log \sum_k c_k\,Q(y|b_k,\epsilon^{-1}A_k)$ exponentiates to

$$\varphi(y) = \sum_k c_k\,Q\big(y\,|\,b_k,\epsilon^{-1}A_k\big) = \sum_k c_k \exp\Big(-\tfrac{1}{2}(y-b_k)^\top \tfrac{A_k}{\epsilon}(y-b_k)\Big),$$

   a sum of bells with centres b_k and precisions A_k/ϵ. This is (21) with

$$\mu_k = b_k, \qquad \Sigma_k^{-1} = A_k/\epsilon, \qquad w_k = c_k\,(2\pi)^{D/2}\det(\Sigma_k)^{1/2},$$

   the last line converting unnormalized bells to normalized densities. Only ratios of the weights matter (an overall factor in φ is absorbed into $\hat{\varphi}$).

3. **Proposition 3.3 is (22).** Substituting $\Sigma_k^{-1} = A_k/\epsilon$ and $\sigma^{-2} = \epsilon^{-1}$ into our S_k, m_k:

$$S_k = (\Sigma_k^{-1} + \sigma^{-2} I)^{-1} = \epsilon\,(A_k + I)^{-1} \qquad \text{(their } \Sigma_n\text{)},$$
$$m_k(x) = S_k(\Sigma_k^{-1}\mu_k + \sigma^{-2}x) = (A_k + I)^{-1}(A_k b_k + x) \qquad \text{(their } \mu_n(x)\text{)}.$$

Their mixture weights are $\gamma_k = \tilde{c}_k / \sum_j \tilde{c}_j$ with

$$\tilde{c}_k = c_k\,(2\pi)^{D/2} \det(S_k)^{1/2}\, Q\big(x \,|\, b_k,\ \epsilon^{-1} I - \epsilon^{-2} S_k\big) \qquad \text{(their } \tilde{w}_n\text{)},$$

and these are our responsibilities $\omega_k(0, x) \propto w_k \mathcal{N}(x; \mu_k, \Sigma_k + \sigma^2 I)$. To see it, note the precision in their Q is

$$\epsilon^{-1} I - \epsilon^{-2} S_k = \epsilon^{-1}(A_k + I)^{-1} A_k = (\epsilon A_k^{-1} + \epsilon I)^{-1} = (\Sigma_k + \sigma^2 I)^{-1},$$

so $Q(x|b_k, \cdot)$ is the unnormalized version of $\mathcal{N}(x; \mu_k, \Sigma_k + \sigma^2 I)$; and

$$\det(S_k)^{1/2} = \epsilon^{D/2} \det(A_k + I)^{-1/2} \propto \det(\Sigma_k)^{1/2} \det(\Sigma_k + \sigma^2 I)^{-1/2}$$

supplies exactly the normalizations that turn c_k into w_k and the bell into a density, up to the common factor $(2\pi\epsilon)^{D/2}$ that cancels in γ_k. Their normalizer Z_x is our $\varphi_0(x)$ up to the same constant. Their appendix derivation (completing the square in the exponent, their (21) and following) is the product rule (1) done by hand.

4. **Corollary 3.5 is the drift of (19).** Write $S_k^t := \epsilon\big((1-t)A_k + I\big)^{-1}$ (their Σ_n^t, with $A_n^t = (1-t)A_n$). Their (16) reads

$$v^*(x, t) = \epsilon \nabla \log \sum_k c_k \det(S_k^t)^{1/2}\, Q\Big(x \,\Big|\, b_k,\ \tfrac{1}{\epsilon(1-t)} I - \tfrac{1}{\epsilon^2(1-t)} S_k^t\Big).$$

The precision simplifies the same way,

$$\tfrac{1}{\epsilon(1-t)}\big(I - ((1-t)A_k + I)^{-1}\big) = \epsilon^{-1}\big((1-t)A_k + I\big)^{-1} A_k = (\Sigma_k + \sigma^2(1-t)I)^{-1},$$

and $\det(S_k^t)^{1/2}$ supplies the normalization, so the sum is $\varphi_t(x)$ of the **Time potential** bullet and $v^* = \sigma^2 \nabla \log \varphi_t = u$. (Their (12) is (18)–(19) with the heat kernel written as $\mathcal{N}(y|x, (1-t)\epsilon I)$.)

Two things in Section 3.3 are *not* just notation. First, they only require the eigenvalues of A_k to lie in $(-1, \infty)$, so a component may have $A_k \preceq 0$: a flat or even growing bell, not a Gaussian. Everything still works because every formula above uses $(A_k + I)^{-1}$, never A_k^{-1}: the heat kernel's precision $\epsilon^{-1} I$ dominates and $S_k \succ 0$. Our Gaussian-mixture φ is the $A_k \succ 0$ sub-case; to cover the rest, read $\Sigma_k^{-1} := A_k/\epsilon$ as the primary object in (22) and never form Σ_k itself. Second, Proposition 3.3 produces *samples* of $P_1 = \pi_1^*$ and nothing else; the paper says explicitly that P_1 is in general not a Gaussian mixture even when P_0 is. Section 10 gives two recipes that close that gap: Recipe A follows the paper and adds one convolution; Recipe B, previewed below, makes everything explicit.

Example 9.2 (One-dimensional worked case)**.** Let $\rho_0 = \mathcal{N}(0, s^2)$ and $\varphi(y) = \frac{1}{2}\mathcal{N}(y; -a, \tau_0^2) + \frac{1}{2}\mathcal{N}(y; a, \tau_0^2)$. Write $\tau^2 = \tau_0^2 + \sigma^2$. Then

$$\varphi_0(x) = \tfrac{1}{2}\mathcal{N}(x; -a, \tau^2) + \tfrac{1}{2}\mathcal{N}(x; a, \tau^2), \qquad \omega_\pm(x) = \frac{\mathcal{N}(x; \pm a, \tau^2)}{\mathcal{N}(x; -a, \tau^2) + \mathcal{N}(x; a, \tau^2)} = \frac{1}{1 + e^{\mp 2ax/\tau^2}},$$

so a particle at $x > 0$ goes to the right branch with probability $\omega_+(x)$ and to the left branch with probability $\omega_-(x) = 1 - \omega_+(x)$ (the "cross-branch" mass). Given the branch, $y \sim \mathcal{N}\big(m_\pm(x), S\big)$ with $S = \big(\tau_0^{-2} + \sigma^{-2}\big)^{-1}$ and $m_\pm(x) = S(\pm a/\tau_0^2 + x/\sigma^2)$. The drift that does the splitting is exact: $u = \sigma^2 \partial_x \log \varphi_t$ with $\varphi_t = \frac{1}{2}\mathcal{N}(x; -a, \tau_0^2 + \sigma^2(1-t)) + \frac{1}{2}\mathcal{N}(x; a, \tau_0^2 + \sigma^2(1-t))$. With $a = 1$, $\tau_0 = 0.5$, $\sigma^2 = 0.1$ the cross-branch probability $\omega_-(x)$ is about 3% at $x = 0.6$ and 0.3% at $x = 1$; shrinking a or raising σ^2 increases it.

The target, however, is *not* closed form here: $\rho_1(y) = \int \rho_0(x)\pi^*(y|x)\,dx$ contains $\omega_\pm(x)$, whose denominator $\varphi_0(x)$ is a sum of two Gaussians and does not cancel against anything. That integral is what Gushchin sample (Prop. 3.3). It is also what `paper_route_sb.py` evaluates, not point by point but as $\rho_1 = \varphi\,\hat{\varphi}_1$ with $\hat{\varphi}_1$ obtained by one Gaussian convolution of $\hat{\varphi} = \rho_0/\varphi_0$ on a fine grid, which is cheap and spectrally accurate; the ring example of `candidate_3d_problems.tex` is built that way with a ρ_0 that is no mixture at all.

The closed-form variant (ours, not the paper's; Recipe B of Section 10). The paper route already gives a usable grid reference with one numerical step (Recipe A). The variant here removes that step and makes ρ_1 explicit, at the price of a mixture ρ_0. The obstruction is the denominator φ_0 in $\hat{\varphi} = \rho_0/\varphi_0$: a Gaussian divided by a sum of Gaussians is not a Gaussian, so $\hat{\varphi}$, $\hat{\varphi}_1$ and $\rho_1 = \varphi\,\hat{\varphi}_1$ are not mixtures. The second Schrödinger equation says ρ_1 is the *product* $\varphi\,\hat{\varphi}_1$, and products of mixtures are mixtures by (1). So the fix is to change which two objects are the inputs. Gushchin choose (ρ_0, φ) and $\hat{\varphi}$ is forced. We choose $(\hat{\varphi}, \varphi)$, both Gaussian, and *both* marginals are outputs. This is legitimate by the converse in Proposition 5.4: $\pi = \hat{\varphi}\,p_1\,\varphi$ is the entropic-optimal plan between its own two marginals, whatever $\hat{\varphi}, \varphi$ are, provided it has mass one. Concretely, keep φ as above and take a single Gaussian

$$\hat{\varphi}(x) = Z^{-1}\mathcal{N}(x; 0, \hat{s}^2), \qquad Z := \mathcal{N}(a; 0, \hat{s}^2 + \tau^2),$$

where the role of Z is only to make the total mass one (step 2 below). Four steps:

1. **Flow φ back to $t = 0$.** By (2) each variance grows by σ^2: $\varphi_0(x) = \frac{1}{2}\mathcal{N}(x; -a, \tau^2) + \frac{1}{2}\mathcal{N}(x; a, \tau^2)$, $\tau^2 = \tau_0^2 + \sigma^2$, as before.

2. **Multiply: $\rho_0 = \hat{\varphi}\,\varphi_0$.** Apply (1) term by term with $A = \hat{s}^2$ (mean 0) and $B = \tau^2$ (mean $\pm a$):

$$\mathcal{N}(x; 0, \hat{s}^2)\,\mathcal{N}(x; \pm a, \tau^2) = \underbrace{\mathcal{N}(\pm a; 0, \hat{s}^2 + \tau^2)}_{= Z \text{ for both signs}}\,\mathcal{N}(x; \bar{x}_\pm, v), \qquad \bar{x}_\pm = \pm\frac{a\,\hat{s}^2}{\hat{s}^2 + \tau^2}, \quad v = \frac{\hat{s}^2\tau^2}{\hat{s}^2 + \tau^2}.$$

   The mean is the precision-weighted average of 0 and $\pm a$; the pull toward the origin is stronger the narrower $\hat{\varphi}$ is. The two constants are equal by symmetry and equal Z by definition, so Z^{-1} cancels them:

$$\rho_0 = \tfrac{1}{2}\mathcal{N}(x; \bar{x}_+, v) + \tfrac{1}{2}\mathcal{N}(x; \bar{x}_-, v),$$

   a probability density.

3. **Flow $\hat{\varphi}$ forward to $t = 1$.** $\hat{\varphi}_1(y) = Z^{-1}\mathcal{N}(y; 0, \hat{s}^2 + \sigma^2)$.

4. **Multiply: $\rho_1 = \varphi\,\hat{\varphi}_1$.** Same rule with variances τ_0^2 and $\hat{s}^2 + \sigma^2$. The constant that appears is $\mathcal{N}(a; 0, \hat{s}^2 + \sigma^2 + \tau_0^2) = Z$, the *same* number, because $\tau^2 = \tau_0^2 + \sigma^2$. This is not a coincidence: both constants are the total mass of the plan, counted once at $t = 0$ and once at $t = 1$. Hence

$$\rho_1 = \tfrac{1}{2}\mathcal{N}(y; \bar{y}_+, v') + \tfrac{1}{2}\mathcal{N}(y; \bar{y}_-, v'), \qquad \bar{y}_\pm = \pm\frac{a(\hat{s}^2 + \sigma^2)}{\hat{s}^2 + \sigma^2 + \tau_0^2}, \quad v' = \frac{(\hat{s}^2 + \sigma^2)\tau_0^2}{\hat{s}^2 + \sigma^2 + \tau_0^2}.$$

The picture: the plan π^* is two bivariate Gaussians, a + branch and a − branch, each of mass exactly $\frac{1}{2}$ at $t = 0$ and $t = 1$ and hence at every t (Section 10). Each branch is a single Gaussian bridge, so ρ_t is always a sum of two Gaussians whose centres slide from $\bar{x}_\pm$ to $\bar{y}_\pm$ and whose variances go from v to v'. At $t = 0$ the narrow $\hat{\varphi}$ pulls the centres together and ρ_0 is one bump; at $t = 1$ the broader $\hat{\varphi}_1$ pulls less and ρ_1 is mildly bimodal. The cross-branch weight $\omega_\pm(x)$ of the paper route still exists but is never needed: it is already summed into the two branch masses. Gushchin's Proposition 3.3 is not replaced, it is bypassed; the paper never needs the density of ρ_1, the grid solver does. With $a = 1$, $\hat{s} = 0.3$, $\tau_0 = 0.5$, $\sigma^2 = 0.1$: ρ_0 has centres ± 0.20 and standard deviation 0.27, a single broad bump; ρ_1 has centres ± 0.43 and standard deviation 0.33, a flat-topped, mildly bimodal density.

Reading Gushchin et al.

Map of the paper's Section 3 (arXiv v2 numbering). 3.1, Theorem 3.1: the construction for a general weak cost. 3.2, Theorem 3.2 with its eq. (14): Proposition 9.1, the conditional plan $\propto \exp((f^* - c)/\epsilon)$, i.e. (10) with $\varphi = e^{f^*/\epsilon}$. 3.3, eq. (15): the LSE potential, our (21); Proposition 3.3: the Gaussian-mixture conditionals, our (22), used to sample P_1. 3.4, Corollary 3.4: $e^{f^*/\epsilon}$ is the Schrödinger potential; Corollary 3.5, eq. (16): the drift, our $u = \sigma^2 \nabla \log \varphi_t$. The dictionary above translates each symbol. Everything from Section 4 on is about the high-dimensional benchmark pairs and neural solvers and can be skipped.

10 Two recipes for an exact grid reference

Why we need a density at all. The paper never evaluates $P_1(y)$ at a point, and never needs to. Its solvers are neural: they take two bags of samples $\{x_i\} \sim P_0$, $\{y_j\} \sim P_1$ as input and return a sampler (a network producing draws from $\hat{\pi}(\cdot|x)$, or a learned drift $\hat{v}$ integrated as an SDE); they are scored by sample-based moment metrics (BW_2^2-UVP). Proposition 3.3, a fast exact sampler, is therefore all they need. Our solver is a grid discretization of (17): its input is ρ_0, ρ_1 as arrays of cell values, since the marginal constraints are imposed pointwise, and its output is ρ_t, m at every cell and time step. Histogramming samples onto the grid would inject Monte Carlo noise of order $N_{\text{cell}}^{-1/2}$ per cell, far above the $O(h^2)$ discretization error the convergence suites are designed to measure; and the error norms of ρ_t and m need the exact ρ_t at every cell. So we need ρ_0, ρ_1, ρ_t and m as functions we can evaluate anywhere, to an accuracy well below 10^{-3}. Two recipes do this. Neither is in the paper as such; the first follows it closely, the second departs in one choice.

What both recipes share. Both rest on one fact, the converse in Proposition 5.4 (the paper's Theorem 3.2):

Take *any* two positive functions $\hat{\varphi}, \varphi$ on $\mathbb{R}^d$ and set $\pi(x, y) = \hat{\varphi}(x)\, \mathcal{K}_\sigma(x, y)\, \varphi(y)$, scaled to total mass one. Then π is the entropic-optimal plan, and the Schrödinger bridge, between its own two marginals. By (18)–(19) the bridge is

$$\rho_t = \varphi_t\, \hat{\varphi}_t, \qquad u = \sigma^2 \nabla \log \varphi_t, \qquad m = \rho_t u,$$

with φ_t the backward and $\hat{\varphi}_t$ the forward heat flow of the two chosen functions. In particular $\rho_0 = \hat{\varphi}\, \varphi_0$ and $\rho_1 = \varphi\, \hat{\varphi}_1$.

So a reference consists of six objects: φ, φ_t, $\hat{\varphi}$, $\hat{\varphi}_t$, ρ_t, u. The two recipes differ only in *which of these are chosen*, *which are forced*, and *how the forced ones are computed*. In both, φ is the Gaussian mixture (21), $\varphi = \sum_k w_k \mathcal{N}(\cdot; \mu_k, \Sigma_k)$, so that φ_t and u are closed form by the **Time potential** and **Drift** bullets of Section 9. Both scripts use the solver's convention $\sigma^2 = 2\varepsilon$: wherever σ^2 appears below the code writes `2*eps`, and the mixture parameters are the arguments `(w, b, Q)` for (w_k, μ_k, Σ_k).

Recipe A: the paper route (`paper_route_sb.py`, class `FixedRho0SB`). This is Gushchin's construction, with the one object they never compute added at the end.

We fix two things. 1. ρ_0: *any* smooth positive density on $[0,1]^d$ that is negligible at the walls, normalized to mass one. It need not be a Gaussian or a mixture; the worked example in `candidate_3d_problems.tex` uses a ring, $\rho_0 \propto \exp(-(|x-c|-r)^2/2w^2)$.

2. $\varphi = \sum_k w_k \mathcal{N}(\cdot; \mu_k, \Sigma_k)$: the centres, widths and weights of the targets, chosen by hand (Section 9). This is the paper's LSE potential $f^* = \sigma^2 \log \varphi$.

Everything else is derived, in this order. 1. $\varphi_t(x) = \sum_k w_k \mathcal{N}\big(x; \mu_k, \Sigma_k + \sigma^2(1-t)I\big)$, by the convolution rule. Closed form. In particular φ_0, which is also the paper's normalizer Z_x up to a constant.

2. $\hat{\varphi} := \rho_0/\varphi_0$. *Forced*: this is the only choice that makes the x-marginal of $\hat{\varphi}\mathcal{K}_\sigma\varphi$ equal to ρ_0 (first Schrödinger equation). Pointwise closed form, but not a mixture: for the ring it is a crescent, the side of the ring facing away from the targets, where φ_0 is smallest. The paper never writes this function down because its sampler never needs it.

3. $\hat{\varphi}_t = \mathcal{N}(0, \sigma^2 t I) * \hat{\varphi}$, the forward heat flow. *The one numerical step.* Evaluate $\hat{\varphi}$ on a fine n_f^d grid; since the solver's cell centres form a tensor grid and the Gaussian kernel is separable, the convolution is $\hat{\varphi}_t = F \times_1 K_x \times_2 K_y (\times_3 K_z)$, with F the fine-grid values and K_x the $n \times n_f$ matrix of one-dimensional Gaussian weights times the fine cell width. Midpoint quadrature of a smooth Gaussian integrand converges spectrally: in the 2D example, $n_f = 256$ is already below 10^{-9} relative to the peak at the earliest solver time, $t = 1/32$. In 3D this is three tensor contractions, $O(n\, n_f^3)$ per time slice, seconds on a GPU.

4. $\rho_t = \varphi_t \hat{\varphi}_t$ and $\rho_1 = \varphi\, \hat{\varphi}_1$. No normalizing constant is needed: ρ_0 has mass one by construction and the Fokker–Planck equation conserves mass. ρ_1 is a product of a mixture and a numerically known function, so it is not explicit, but it is known at every cell to quadrature accuracy, which is all the solver needs.

5. $u = \sigma^2 \nabla \log \varphi_t$ from step 1, closed form and independent of ρ_0; $m = \rho_t u$.

6. (Optional.) The static plan $\pi^*(y|x) = \varphi(y)\mathcal{N}(y; x, \sigma^2 I)/\varphi_0(x)$, which is the paper's Proposition 3.3 mixture (22).

What is exact and what is not. Steps 1, 2, 5, 6 are closed form. Step 3 carries quadrature error, below 10^{-9} for $n_f \geq 256$ in the example. Step 4 inherits it. The only other error is the $\mathbb{R}^d$-versus-box mismatch, bounded by the mass of ρ_t outside the walls ($8 \cdot 10^{-7}$ in the ring example). Both are orders of magnitude below the solver's $O(h^2)$ error.

Gain and loss. Gain: complete freedom in ρ_0. Loss: ρ_1 has no formula, and there is no branch-mass table. Two practical rules: keep ρ_0 smooth on the scale $\sqrt{\sigma^2\, dt}$, the blur at the first time step, or $\hat{\varphi}$ in step 2 is rough; and choose φ so that φ_0 does not vary by orders of

magnitude across the support of ρ_0, or $\hat{\varphi}$ concentrates on a sliver of it (the ring-to-crescent effect, factor 261 in the example).

Recipe B: both potentials as mixtures (`mixture_sb.py`, class `MixtureSB`). The same construction with step 2 of Recipe A replaced by a choice, which removes the numerical step and makes everything explicit.

We fix two things, neither of which is a marginal.
1. $\hat{\varphi} = \sum_i u_i \mathcal{N}(\cdot; a_i, P_i)$: a Gaussian mixture for the *initial* potential, with hand-picked (u_i, a_i, P_i), the code's `(u, a, P)`. This replaces "fix ρ_0". ρ_0 is now an output.
2. $\varphi = \sum_k w_k \mathcal{N}(\cdot; \mu_k, \Sigma_k)$ as in Recipe A.

Everything else is derived.
1. $\varphi_t(x) = \sum_k w_k \mathcal{N}\big(x; \mu_k, \Sigma_k + \sigma^2(1-t)I\big)$, as before.
2. $\hat{\varphi}_t(x) = \sum_i u_i \mathcal{N}\big(x; a_i, P_i + \sigma^2 t I\big)$, by the convolution rule. This is Recipe A's step 3, now closed form because $\hat{\varphi}$ is a mixture.
3. Mass. The product $\hat{\varphi}\mathcal{K}_\sigma\varphi$ has total mass $K = \sum_{i,k} u_i w_k \mathcal{N}\big(a_i; \mu_k, P_i + \Sigma_k + \sigma^2 I\big)$; divide by it. The normalized summands
$$\Omega_{ik} = u_i w_k \mathcal{N}\big(a_i; \mu_k, P_i + \Sigma_k + \sigma^2 I\big)/K, \qquad \sum_{i,k} \Omega_{ik} = 1,$$
are the *branch masses*: Ω_{ik} is the mass of the plan flowing from component i of $\hat{\varphi}$ to component k of φ. They do not depend on t.
4. $\rho_t = \varphi_t \hat{\varphi}_t / K$. Each term is a product of two Gaussians in the same variable, so by (1) with $A = P_i + \sigma^2 t I$ and $B = \Sigma_k + \sigma^2(1-t)I$,
$$\rho_t = \sum_{i,k} \Omega_{ik}\, \mathcal{N}\big(\cdot;\, c_{ik}(t),\, C_{ik}(t)\big), \tag{23}$$
with $C_{ik}(t) = \big(A^{-1} + B^{-1}\big)^{-1}$ and $c_{ik}(t) = C_{ik}(t)\big(A^{-1}a_i + B^{-1}\mu_k\big)$. The constants $\mathcal{N}(a_i; \mu_k, A + B)$ produced by the product rule are exactly the numerators of Ω_{ik}, since $A + B = P_i + \Sigma_k + \sigma^2 I$ for every t. This is why the weights are t-independent. ρ_0 and ρ_1 are (23) at $t = 0$ and $t = 1$: both are mixtures of $I \cdot K$ Gaussians with the same weights.
5. $u = \sigma^2 \nabla \log \varphi_t$, $m = \rho_t u$, as before.
6. (Optional.) The static plan $\pi^*(y|x) = \varphi(y)\mathcal{N}(y; x, \sigma^2 I)/\varphi_0(x)$, unchanged, and the joint plan $\pi^*(x, y) = \sum_{i,k} \Omega_{ik} \times$ one bivariate Gaussian on $\mathbb{R}^{2d}$ per branch.

Why it all closes. For one pair (i, k),
$$\hat{\varphi}_i(x)\, \mathcal{K}_\sigma(x, y)\, \varphi_k(y) \;\propto\; \exp\Big(-\tfrac{1}{2}\Big[(x - a_i)^\top P_i^{-1}(x - a_i) + \tfrac{|x-y|^2}{\sigma^2} + (y - \mu_k)^\top \Sigma_k^{-1}(y - \mu_k)\Big]\Big),$$
whose exponent is one quadratic in the stacked vector $(x, y) \in \mathbb{R}^{2d}$. So each branch is a single Gaussian on $\mathbb{R}^{2d}$, the plan is a Gaussian mixture on $\mathbb{R}^{2d}$, and every marginal in x, in y, or at any intermediate time is a Gaussian mixture with the same weights. Example 9.2 is the $I = 1$, $K = 2$ case.

What is exact. Everything; nothing is sampled or integrated. Checks recorded in the code and in `candidate_3d_problems.tex`: (23) equals the product $\varphi_t\hat{\varphi}_t/K$ to 10^{-15}; ρ_1 equals $\int \rho_0 \pi^*(\cdot|x)\, dx$ by quadrature to $3 \cdot 10^{-13}$; the finite-difference Fokker–Planck residual decays as $O(h^2)$; and $I = K = 1$ reproduces the Gaussian closed form of Bunne et al.

Gain and loss. Gain: no numerical step, explicit ρ_1, and a branch-mass table Ω_{ik} that says in advance how much mass goes where. Loss: ρ_0 is a mixture of $I \cdot K$ Gaussians, not a free choice; a single-Gaussian ρ_0 with a multimodal ρ_1 is impossible here unless $K = 1$. In the 1D example the two components of ρ_0 sit so close that it looks like one bump.

Side by side.

	Recipe A (paper route)	Recipe B (both mixtures)
chosen	ρ_0 (anything smooth), φ (mixture)	$\hat{\varphi}$ (mixture), φ (mixture)
forced	$\hat{\varphi} = \rho_0/\varphi_0$	$\rho_0 = \hat{\varphi}\varphi_0/K$
$\hat{\varphi}_t$	one Gaussian convolution, $\leq 10^{-9}$	closed form
ρ_1	numerical product, no formula	mixture, (23) at $t = 1$
ρ_t, u, m	product and closed form	closed form
branch masses	not available	Ω_{ik}, explicit
use when	a specific non-mixture ρ_0 is wanted	an all-explicit multimodal test is wanted

What the solver is compared against. In the solver's convention, with $\varepsilon = \sigma^2/2$:

- ρ_t on cell centres, from (23) (B) or from step 4 (A);
- $m = 2\varepsilon\,\hat{\varphi}_t \nabla\varphi_t$ on the staggered faces (`staggered_reference` in both classes evaluates at exactly the solver's points, so no interpolation);
- the cost $\int_0^1 \int |m|^2/2\rho$, which by (16) equals $\sigma^2\,\mathrm{KL}(\mathbb{P}^*\|\mathbb{W}^\sigma)$; it is evaluated by quadrature of the reference on a fine grid, not by a formula of the Bunne type.

What this can and cannot do, and what is the paper's. The pair (ρ_0, ρ_1) produced by Recipe B is always two Gaussian mixtures, but a *derived* pair: shared weights Ω_{ik}, means and covariances tied by (23). Given two arbitrary Gaussian mixtures, the bridge between them is not closed form (the potentials are not mixtures), and neither recipe solves it. Both are generators of test pairs with an exact solution, which is what a convergence study needs and what the paper itself set out to build. From the paper we use three things: the idea of choosing the potential and reading off the pair, which is what gives a multimodal bridge with an exact answer at all (the single-Gaussian references of Bunne and Janati are the $I = K = 1$ corner); Theorem 3.2 as the citable optimality and uniqueness statement (the underlying fact is the classical Schrödinger system); and the LSE formulas of Proposition 3.3 and Corollary 3.5, which after the dictionary are (22) and the drift we evaluate. We do not use the sampler, Theorem 3.1, or the benchmark pairs. Recipe A adds the forward heat flow of $\hat{\varphi}$, which the paper never needs; Recipe B adds the choice of $\hat{\varphi}$ as a mixture. Theorem 3.2 never required P_0 to be prescribed; the paper fixed it because they wanted a given dataset or a clean Gaussian at $t = 0$. Both recipes could be derived from the Schrödinger system and the two Gaussian identities alone; citing the paper buys the recognized framing and the theorem.

Boundary conditions. Every formula above is on $\mathbb{R}^d$. The solver lives on $[0,1]^d$ with no-flux walls, whose reference process is *reflected* Brownian motion and whose heat kernel is the Neumann kernel, a cosine series rather than (14). The references remain valid as long as ρ_t carries negligible mass at the walls for all t (tail mass $\lesssim 10^{-7}$ in the examples), because then the two kernels agree to that accuracy on the support. The same caveat applies to the Gaussian closed form of Bunne et al.

11 Symbol table and reading map

Object	This primer	Gushchin et al. (check pdf)	Our code
noise of reference process	σ^2	ϵ	2ε
Fokker–Planck diffusion	$\sigma^2/2$	$\epsilon/2$	ε
EOT regularization, cost $\frac{1}{2}\lvert x-y\rvert^2$	σ^2	ϵ	2ε
EOT regularization, cost $\lvert x-y\rvert^2$ (Janati)	$2\sigma^2$	–	4ε
source, target	ρ_0, ρ_1	P, Q	`rho0, rho1`
plan, conditionals	π^*, $\pi^*(\cdot\lvert x)$	π^*, $\pi^*(\cdot\lvert x)$	static plan
dual potential	$f = \epsilon \log \varphi$ (unused)	f^*	–
mixture component	$\mathcal{N}(\cdot; \mu_k, \Sigma_k)$, weight w_k	$Q(\cdot\lvert b_n, \epsilon^{-1}A_n)$, weight w_n ($= c_k$ in the dictionary)	`(w,b,Q)`, `(u,a,P)`
conditional covariance, mean	S_k, $m_k(x)$	$\Sigma_n = \epsilon(A_n + I)^{-1}$, $\mu_n(x)$	–
responsibilities	$\omega_k(t, x)$	γ_n (at $t = 0$)	–
potentials	$\hat\varphi, \varphi$	$\tilde\varphi^*$, $\varphi^* = e^{f^*/\epsilon}$	`phihat, phi`
heat / entropic kernel	$p_1 = \mathcal{N}(y; x, \sigma^2 I)$	$\exp(-\lvert x-y\rvert^2/2\epsilon)$	–
time-t marginal	$\rho_t = \varphi_t\hat\varphi_t$	same	`rho(t)` on centres
drift, momentum	$u = \sigma^2\nabla \log \varphi_t$, $m = \rho u$	$\epsilon\nabla \log \varphi_t$	$m = 2\varepsilon\hat\varphi_t\nabla\varphi_t$ on faces
dynamic cost	$\iint \lvert m\rvert^2/2\rho = \sigma^2 \mathrm{KL}(\mathbb{P}^*\lVert\mathbb{W}^\sigma)$	$\epsilon\ \mathrm{KL}(\cdot \,\lVert\, W^\epsilon)$	LADMM objective

Reading map for Gushchin et al., Sections 2–3.

1. Section 2, entropic OT and weak OT: Sections 3–6 here. Goal: recognize (7), (8) and the pointwise condition (13).
2. Section 2, dynamic Schrödinger bridge: Sections 7–8. Goal: $\rho_t = \varphi_t\hat\varphi_t$ and drift $\epsilon\nabla \log \varphi_t$.
3. Section 3.1, Theorem 3.2 and the drift display: Proposition 9.1.
4. Proposition 3.3 and Corollary 3.5: (22) and Example 9.2.
5. Stop there. Sections 4–6 are the benchmark and neural solvers.

Further reading, in order of usefulness. Léonard, *A survey of the Schrödinger problem and some of its connections with optimal transport*, DCDS-A 34 (2014), arXiv:1308.0215, Sections 1–3, for the dynamic/static reduction and the potentials; Peyré & Cuturi, *Computational Optimal Transport*, Chapter 4, for entropic OT and Sinkhorn; Chen, Georgiou, Pavon, *Stochastic control liaisons*, SIAM Review 63 (2021), for the control form (16); Bunne et al., AISTATS 2023, Theorem 3, and Janati et al., NeurIPS 2020, Theorem 1, for the Gaussian closed forms used in Section 5.